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Windows Forensic Analysis Playbook

**A MICROLEARNING SERIES
OF MODERN INVESTIGATIONS**

**6 Critical Windows Artifacts
Every Investigator Should Know**



Windows Forensic Analysis Playbook

A MICROLEARNING SERIES OF MODERN INVESTIGATIONS

This playbook highlights six critical Windows artifacts investigators rely on during analysis. It explains what each artifact is, where it is stored, and what it reveals, providing a practical reference for consistent investigations in Windows environments.

Artifacts covered:

- ▶ Jump Lists
- ▶ LNK Files
- ▶ RecentDocs/OpenSave MRU
- ▶ Prefetch
- ▶ SRUM
- ▶ Windows Event Logs

These are not exhaustive deep dives. They are practical, high-value reference guides that support investigations, rapid response scenarios, and day-to-day forensic work.

Jump Lists

ARTIFACT QUICK REFERENCE



What Are Jump Lists?

Jump Lists were introduced in Windows 7 and are accessed through the taskbar by right-clicking an icon. They track up to ~2,000 recently or frequently accessed files per applications by a user account, with each Jump List entry recording metadata for the file that was accessed, including timestamps and the volume it resided on, making it a reliable source of evidence for user activity and context.



Why Do They Matter in an Investigation?

- **Why investigators care:** Jump Lists provide evidence of user account activity in opening or interacting with specific files and applications.
- **How much data:** They track up to ~2,000 recently or frequently accessed files, per application, by a given user account.
- **Caveat:** Jump Lists are created the first time applications are used by that user account, so they cannot be used to determine applications that are pinned to the taskbar.



Storage Location

Jump List data is stored inside each user profile at:

```
C:\Users\<username>\AppData\Roaming\Microsoft\Windows\Recent\CustomDestinations
```

```
C:\Users\<user>\AppData\Roaming\Microsoft\Windows\Recent\CustomDestinations\
```

- **Access method:** Extract the files from this location and parse them with forensic tools, or parse these folders directly.
- **Contents:** Each Automatic Jump List contains recent activity for an application. Each Custom Jump List stores data about tasks and actions the application developer wished to highlight to the user. Each Jump List is named for the globally unique Application ID corresponding to that application. Each Jump List entry is a shell item and contains metadata about the file that was accessed.



How to Use in Investigations

Jump Lists are excellent for reconstructing user behavior and demonstrating the application they used. They provide forensic examiners with a record of files accessed and when. Correlating Jump Lists with Prefetch, LNK shortcuts, RecentDocs, or Shellbags strengthens activity timelines and builds robust evidence.

Use Cases

- **User Activity Attribution**
It connects users to files they accessed via applications.
- **Behavior Reconstruction**
This maps app usage and file interaction over time.
- **Example Case**
This data can show that an employee opened confidential spreadsheets prior to leaving the company.



Tips

- **Timestamps:** Records the last time files were accessed as well as the target file timestamps when they were accessed
- **Evidence of Program Execution:** Jump Lists are created the first time an application executes, so provides evidence of program execution
- **File Access:** Demonstrates which files a user accessed recently
- **Correlation:** Best when validated with Prefetch, LNK files and MRU artifacts



Quick Commands (Forensic Parsing)

A common tool for parsing Jump Lists is **JLECmd.exe**, which extracts and formats the data for analysis.

<https://github.com/EricZimmerman/JLECmd>

Example command:

```
JLECmd.exe -d AutomaticDestinations  
--csv output-folder --csvf JumpLists.csv
```

- `-d AutomaticDestinations` tells the tool to process all Jump List files in the directory.
- `--csv output-folder` exports the parsed results to output-folder.
- `--csvf JumpLists.csv` exports the parsed results to a CSV file called JumpLists.csv.
- The CSV output file will include accessed file names, paths, volume information, and timestamps for when the files were last accessed by the user account.



Jump List Parsing Tools

- Jump List parsing with JumpList Explorer
<https://ericzimmerman.github.io/#!index.md>
GUI-based tool
- Jump List parsing with JLECmd
<https://github.com/EricZimmerman/JLECmd>

```
JLECmd.exe -d AutomaticDestinations  
--csv output-folder --csvf JumpLists.csv
```
- Jump List parsing with TZWorks jmp
https://tzworks.com/prototype_page.php?proto_id=20

```
dir AutomaticDestinations /b /s | jmp64  
-pipe -csv > results.csv
```
- Using Velociraptor's Jump List parser
https://docs.velociraptor.app/artifact_references/pages/windows.forensics.jumplist

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LNK Files

ARTIFACT QUICK REFERENCE



What Are LNK Shortcuts?

Shortcut (LNK) files are automatically generated by Windows whenever a user creates or opens a file or folder. These files contain metadata such as the file path, timestamps, and volume or network share information of the created or opened file/folder. The files themselves also have their own filesystem metadata, which indicates when a file or folder at that path was first and last accessed.



Why Do They Matter in an Investigation?

- **Why investigators care:** Even if the original file has been deleted, LNK files may persist and provide evidence of file access or staging. This makes them powerful for proving existence of sensitive files.
- **How much data:** The Recent folder shows approx. 149 LNK files to the user in Explorer, but many more may be present in this folder and accessible using the command line or forensic tools.
- **Caveat:** LNK files are created when a file or folder is created or opened, alongside LNK files for parent or grandparent folders. Therefore, the existence of a LNK file in itself does not indicate that a file was actually opened, or whether the contents were modified. They must be interpreted alongside other artifacts.



Storage Location

LNK files are typically stored in the user profile at:

```
C:\Users\<user>\AppData\Roaming\Microsoft\Windows\Recent\
```

- **Access method:** Extract the files from this location and parse them with forensic tools, or parse these files directly.
- **Contents:** Target file path, timestamps (creation, access, modification), file size, and volume or network share information, including system identifiers such as volume serial numbers.



How to Use in Investigations

LNK files provide residual evidence of user interaction with files, even when the original files are gone. They are excellent for confirming the presence of sensitive documents, uncovering insider threats, and reconstructing how files moved between systems. Correlate with MRU lists and Jump Lists for stronger timelines.

Use Cases

- **File Access Proof**
It demonstrates that sensitive files were present on the system and either created or accessed by the user account.
- **Example Case**
This data can show that “Payroll2025.xlsx” was opened from a USB drive and provide the volume serial number for that volume, to allow correlation with seized media.



Tips

- **Metadata Rich:** Records target paths, timestamps, and system info
- **Exfil Clues:** Useful for spotting staged data before exfiltration
- **Persistence:** Survives even after files are deleted
- **Attribution:** Can link access to specific removable drives or network shares



Quick Commands (Forensic Parsing)

A common tool for parsing LNK files is **LECcmd.exe**, which extracts and formats the contents of LNK files.

<https://github.com/EricZimmerman/LECcmd>

Example command:

```
LECcmd.exe -d "%AppData%\Microsoft\Windows\Recent" --csv output-folder --csvf LNK.csv
```

- `-d "%AppData%\Microsoft\Windows\Recent"` tells the tool to process all Jump List files in the directory.
- `--csv output-folder` exports the parsed results to output-folder.
- `--csvf LNK.csv` exports the parsed results to a CSV file called LNK.csv.



LNK Parsing Tools

- LNK file parsing with LECmd
<https://github.com/EricZimmerman/LECcmd>

```
LECcmd.exe -d "%AppData%\Microsoft\Windows\Recent" --csv output-folder --csvf LNK.csv
```
- LNK file parsing with TZWorks lp
https://tzworks.com/prototype_page.php?proto_id=11

```
dir "%AppData%\Microsoft\Windows\Recent" /b /s | lp -pipe -csv > results.csv
```
- Using Velociraptor's LNK file parser
https://docs.velociraptor.app/artifact_references/pages/windows.forensics.lnk

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RecentDocs/OpenSave MRU

ARTIFACT QUICK REFERENCE



What Are Recent/Docs/OpenSave MRU?

The RecentDocs and OpenSave MRU (Most Recently Used) keys in the Windows Registry track the most recent files and folders opened by a user, and the order in which they were accessed. These artifacts store file names, extensions, and sometimes directory paths, allowing investigators to reconstruct recent user activity.

RecentDocs data is used to populate places like the “Recent” menus present in some Start menus. OpenSave MRU data tracks files that have been opened or saved within a Windows common dialog box.



Why Do They Matter in an Investigation?

- **Why investigators care:**

They provide visibility into which files were accessed and in what order. This is especially useful in insider threat or intellectual property theft cases.

- **How much data:**

- RecentDocs stores 150 of the most recent items accessed by the user account within the parent key, with up to 20 of each extension (30 for folders or files without an extension) in subkeys.
- OpenSavePidlMRU stores the most recently accessed 20 files of any extension in the * key and another 20 of each extension in subkeys.

- **Caveat:**

Both keys only show that files were opened; they don't indicate how the files were used or whether they were modified.



Storage Location

These MRU keys are stored inside the NTUSER.DAT registry hive for each user account:

```
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs
```

```
NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSavePidlMRU
```

- **Access method:** Extract the NTUSER.DAT from each user's profile folder and parse them with forensic tools, or parse these files directly.
- **Contents:** File names, extensions, folders, and MRU lists that record the sequence of recently accessed files.



How to Use in Investigations

RecentDocs and Open/Save MRU keys are best used to uncover recent file access patterns, hidden, deleted, or unusual directories, and user habits. They provide evidence of what a user interacted with, the order items were accessed, and to some extent when, as the most recently accessed item can be correlated with a timestamp. Pairing this data with Jump Lists and LNK files confirms user activity.

Use Cases

- **Hidden File Access**

It highlights files and directories a user accessed that may not be obvious elsewhere.

- **Example Case**

This data can show that a departing employee accessed sensitive files before resignation.



Tips

- **Dialog-Based Activity:** Captures files opened via common Open/Save dialogs used by applications
- **Usefulness:** Excellent for insider threat and fraud cases, to show user interaction with files and folders
- **Temporal Ordering:** MRU lists show order of access
- **Extension-Specific Tracking:** Stores the last 20 files accessed of each extension type in each set of subkeys



Quick Commands (Forensic Parsing)

A common tool for parsing MRU Registry keys is **KAPE**, which extracts and formats the data for analysis. Both GUI and command line versions of the tool are available.

<https://www.kroll.com/en/services/cyber/incident-response-recovery/kroll-artifact-parser-and-extractor-kape>

Example command:

```
.\kape.exe --msource source-volume --mdest output-folder --module RECcmd_UserActivity --mef csv
```

- `--msource source-volume` tells the tool to process evidence in this volume.
- `--mdest output-folder` exports the parsed results to output-folder.
- `--module RECcmd_UserActivity` runs the built-in 'RECcmd_UserActivity' module.
- `--mef csv` exports the parsed results to a CSV file.
- The CSV output file will include parsed RecentDocs and OpenSavePidlMRU key data, as well as other artifacts indicating user activity.



RecentDocs/OpenSave MRU Parsing Tools

- MRU key parsing with Registry Explorer <https://ericzimmerman.github.io/#index.md>
GUI-based tool, with built-in plugins for these MRU keys. Simply navigate to a key for the relevant plugin to parse the data. A command-line version of the tool called RECcmd is also available.
- Recent user activity parsing with KAPE <https://www.kroll.com/en/services/cyber/incident-response-recovery/kroll-artifact-parser-and-extractor-kape>

```
.\kape.exe --msource source-volume --mdest output-folder --module RECcmd_UserActivity --mef csv
```
- Using Velociraptor's RecentDocs parser <https://docs.velociraptor.app/artifact-references/pages/windows.registry.recentdocs>

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Windows Event Logs

ARTIFACT QUICK REFERENCE



What Are Windows Event Logs?

Windows Event Logs (EVTX files) record detailed system, application, and user account activity. These logs include user logon attempts, service creation, network connections, and many other events. They are one of the richest sources of forensic data, and provide essential visibility into attacker persistence and actions on a system.

Since these files are intended to provide key information to system administrators in order to help identify and diagnose issues, they are also very well-documented, which is rare for Windows artifacts.



Why Do They Matter in an Investigation?

- **Why investigators care:** Event Logs allow analysts to attribute actions to specific applications, system processes, or user accounts and facilitate the building of timelines for attacker behavior, including authentication, persistence, and lateral movement.
- **How much data:** There are three main event logs: System, Security, and Application, as well as hundreds of custom logs. The Security log is typically the most valuable to an investigation. The volume of data stored in these logs depends on the system's configuration and available space.
- **Caveat:** Logs can be extremely verbose, and investigators must filter for the most relevant Event IDs. Overlooked or improperly interpreted entries may cause missed evidence. By default, a lot of useful events are disabled. Logs are cyclical, so enabling too many events will result in logging for only a short timeframe.



Storage Location

Windows Event Logs are stored at:

```
C:\Windows\System32\winevt\Logs\*.evtx
```

- **Access method:** Extract EVTX files from the system and parse them with forensic tools, or parse these files directly.
- **Contents:** Each log file contains thousands of events, including authentication attempts, process and service creation, and system errors.



How to Use in Investigations

Event Logs are indispensable for understanding what happened on a system, which can include attacker activity. They provide attribution to user accounts, can document persistence mechanisms and show when attackers moved laterally. To maximize value, investigators should focus on high-value Event IDs (e.g., 4624/4625 logons, 4698 task creation, 7045 service installation) and correlate with file system and registry artifacts. If there is an opportunity to configure event logging on a system prior to an investigation, enable additional events such as failed logons, task and service creation, and RDP connections.

Use Cases

- **Persistence Detection**
Identify malicious tasks or services created by attackers.
- **Example Case**
This data can be used to show that Event ID 7045 recorded the installation of a malicious service.



Tips

- **Execution Evidence:** Can capture persistence events such as tasks and service creation, although these are not enabled by default
- **Authentication Traces:** Can log both successful and failed logon attempts. Failed logons are not logged by default
- **Lateral Movement:** Can log remote connections, both inbound and outbound, although this is not enabled by default
- **Filtering Need:** Must be carefully filtered to identify attacker TTPs
- **Breadth:** Covers logons, service activity, system events, and user account activity



Quick Commands (Forensic Parsing)

A common tool for viewing event logs is **Event Log Explorer**, which is a GUI-based tool.

Event Logs can be merged, searched, and filtered. Custom columns can also be used to parse event fields of interest for easy viewing. Events can be filtered by Event ID, user account, system, keywords, or other fields to highlight attacker activity. Data can be exported to XLSX, PDF, HTML, and other formats.

<https://www.eventlogxp.com>



Windows Event Log Parsing Tools

- Event Log parsing with Event Log Explorer
<https://www.eventlogxp.com>
GUI-based tool
- Event Log parsing with EvtxECmd
<https://github.com/EricZimmerman/evtexecmd>

```
EvtxECmd.exe -d Logs --csv output-folder --csvf EventLogs.csv
```
- Parse specific Event Logs with Velociraptor
https://docs.velociraptor.app/docs/forensic/event_logs

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Prefetch

ARTIFACT QUICK REFERENCE



What Is Prefetch?

Prefetch files are created by Windows to speed up application launching. Each .pf file records information about the executable that ran, including the last eight times it was executed, the total run count, and associated DLLs and files the executable used within the first few seconds the last time it ran.



Why Do They Matter in an Investigation?

- **Why investigators care:** Prefetch provides some of the strongest evidence of program execution. It shows not only whether a file ran, but also how often and when.
- **How much data:** There can be up to 1,024 .pf files in the Prefetch folder.
- **Caveat:** On servers, Prefetch is often disabled by default.



Storage Location

Prefetch data is stored at:

`C:\Windows\Prefetch\*.pf`

- **Access method:** Extract PF files from the system and parse them with forensic tools, or parse these files directly.
- **Contents:** Executable name and path, run count, last eight execution times, and list of DLLs and files loaded the last time the executable ran. The first execution time can also be estimated based on the creation time of the PF file minus up to 10 seconds.



How to Use in Investigations

Prefetch is critical for proving program execution, validating incident timelines, and correlating malware execution with user activity. Combined with SRUM, BAM/DAM, and UserAssist data, it builds a comprehensive execution story.

Use Cases

- **Execution Confirmation**
This proves whether malware executed.
- **Example Case**
This data can validate that “encryptor.exe” ran during a ransomware incident.



Tips

- **Execution Proof:** Shows that a program ran on the system
- **Timeline Anchors:** Captures first (approx.) and last eight run times
- **Run Frequency:** Records how many times it executed
- **Context:** Lists DLLs and file handles accessed the last time it ran



Quick Commands (Forensic Parsing)

A common tool for Prefetch parsing is **PECmd.exe**, which extracts and formats the contents of Prefetch files into a CSV file.

<https://github.com/EricZimmerman/PECmd>

Example command:

```
PECmd.exe -d "C:\Windows\Prefetch" --csv output-folder  
--csvf prefetch.csv
```

- `-d "C:\Windows\Prefetch"` tells the tool to process all Prefetch files in the directory.
- `--csv output-folder` exports the parsed results to output-folder
- `--csvf prefetch.csv` exports the parsed results to a CSV file called prefetch.csv.



Prefetch Parsing Tools

- Prefetch Parsing with PECmd
<https://github.com/EricZimmerman/PECmd>

```
PECmd.exe -d "C:\Windows\Prefetch"  
--csv output-folder --csvf prefetch.csv
```
- Prefetch parsing with WinPrefetchView
https://www.nirsoft.net/utills/win_prefetch_view.html
GUI-based tool
- Prefetch parsing with TZWorks Windows Prefetch Parser (pf)
https://tzworks.com/prototype_page.php?proto_id=1

```
dir "C:\Windows\Prefetch" /b /s |  
pf64 -pipe -csv > results.csv
```
- Using Velociraptor's Prefetch parser
https://docs.velociraptor.app/artifact_references/pages/windows.forensics.prefetch/

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SRUM

ARTIFACT QUICK REFERENCE



What Is SRUM?

The System Resource Usage Monitor (SRUM) is a Windows feature introduced in Windows 8 and enabled by default, that tracks system and application activity in approximately hourly increments. Data within this artifact is used to populate Task Manager. The SRUM database logs details about applications launched, user account activity, and network usage.



Why Do They Matter in an Investigation?

- **Why investigators care:** SRUM provides visibility into what applications were running, which user accounts executed them, and how much data they sent/received across the network as well as which network the system was connected to at that time. This is critical information in modern investigations for detecting exfiltration and correlating user behavior.
- **Caveat:** Since data is only retained for up to 60 days, the usefulness of SRUM diminishes as time passes after an incident. Entries are dumped into the database in periodic blocks, so the exact time activity occurred cannot be established and this can potentially be a point of confusion. The SRUDB.dat is not always written on shutdown, so entries may be missed.
- **How much data:** SRUM only retains data for about 30–60 days.



Storage Location

SRUM data is stored in the following database file:

```
C:\Windows\System32\SRU\SRUDB.dat
```

Database table names are stored in the following Registry key:

```
SOFTWARE\Microsoft\Windows NT\CurrentVersion\SRUM\Extensions
```

- **Access method:** Extract the SRUDB.dat file from the system and parse it offline with forensic tools, or parse it directly.
- **Contents:** Application execution logs, user account attribution, network connection information, and per-app network usage.



How to Use in Investigations

SRUM is valuable for reconstructing recent activity during compromises, especially for identifying applications used in exfiltration. It allows investigators to tie application launches to user accounts and correlate them with network traffic. Always collect SRUM data quickly, as its retention is limited.

Use Cases

- **Application + Network Correlation**
They connect apps to suspicious data transfers.
- **Example Case**
This data can detect WinSCP sending large amounts of data during a suspected insider exfiltration.



Tips

- **Usage Tracking:** Logs applications launched per approx. hour
- **User Attribution:** Connects application use to user accounts
- **Network Monitoring:** Tracks bytes sent/received by application
- **Retention:** Only stores ~30–60 days of history



Quick Commands (Forensic Parsing)

A common tool for parsing SRUM is **srum-dump**, which converts the database into CSV output, as well as correlating data with the Registry to convert identifiers to the data they represent.

<https://github.com/MarkBaggett/srum-dump>

Example command:

```
srum_dump.exe -i SRUDB.dat -o output-folder  
-r SOFTWARE -f xls
```

- `-i SRUDB.dat` specifies the SRUM database file to parse.
- `-o output-folder` outputs the parsed data into a folder called output-folder.
- `-r SOFTWARE` parses the Registry hive called SOFTWARE to correlate identifiers.
- `-f xls` exports the parsed results to an XLS file.



SRUM Parsing Tools

- SRUM Parsing with **srum-dump**
<https://github.com/MarkBaggett/srum-dump>

```
srum_dump.exe -i SRUDB.dat  
-o output-folder -r SOFTWARE -f xls
```
- A GUI-based **srum-dump** tool is also available.
- SRUM Parsing with **SRUMECmd**
<https://github.com/EricZimmerman/Srum>

```
SrumECmd.exe -f SRUDB.dat -r  
SOFTWARE --csv output-folder
```
- Using Velociraptor's SRUM parser
https://docs.velociraptor.app/artifact_references/pages/windows.forensics.srum/

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